

Two-dimensional conformal field theory with affine Lie algebra symmetry

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Motivation

- String theories on Lie group target spaces lead to affine Lie algebras at non-negative integer levels.
- Fractional level affine $\mathfrak{sl}(2)$ was proposed by Kent as a means to a uniform construction of the Virasoro minimal models via cosets.
- Virasoro minimal models can also be constructed from the same fractional levels by quantum Drinfeld-Sokolov reduction.
- Kac and Wakimoto showed that for precisely these fractional levels there exist highest weight modules whose characters have nice modular properties. This generalises to affinisations of all finite-dimensional simple Lie algebras.

Conformal field theory from representation theory

One constructivist approach to conformal field theory is the following:

- ❶ Pick a vertex operator algebra V (a.k.a. chiral algebra or holomorphic fields)
- ❷ Find a module category of V that satisfies:
 - ▶ Every module has a well defined character.
 - ▶ The set of characters closes under modular transformations.
 - ▶ The category closes under the fusion of modules.
- ❸ Assemble everything to a consistent conformal field theory (3-pt constants, modular invariants, sewing constraints).

Today's topic: Point 2 for vertex operator algebras constructed from affine Lie algebras.

Affine Lie algebras

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra. As a vector space its affinisation is

$$\widehat{\mathfrak{g}} = \mathfrak{g} \otimes \mathbb{C}[t, t^{-1}] \oplus \mathbb{C}K,$$

where K is central. For $x, y \in \mathfrak{g}$, the $\widehat{\mathfrak{g}}$ -Lie bracket is given by

$$[x \otimes t^m, y \otimes t^n] = [x, y] \otimes t^{m+n} + \delta_{m+n,0} \cdot m \cdot (x, y) \cdot K,$$

where $(\cdot, \cdot)$ is the Killing form normalised such that the longest root has length $\sqrt{2}$.

We fix K to act as some constant multiple of the identity, $K = k\mathbf{1}$, on modules and define $x_m = x \otimes t^m$. The factor k is called the level.

For notational simplicity we will specialise to $\mathfrak{sl}(2)$ for most of the rest.

Ordinary $\mathfrak{sl}(2)$

Denote the standard basis of $\mathfrak{sl}(2)$ by e, h, f . The Lie bracket is

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h,$$

and the standard normalisation of the Killing form is

$$(h, h) = 2, \quad (e, f) = 1.$$

The quadratic Casimir

$$Q = \frac{1}{2}h^2 + ef + fe$$

generates the centre of the universal enveloping algebra: it commutes with all generators of $\mathfrak{sl}(2)$. On simple modules $Q = \Delta \cdot \mathbf{1}$, $\Delta \in \mathbb{C}$ and Δ is important for characterising such modules.

Proposition [Gabriel]

The simple $\mathfrak{sl}(2)$ weight modules (= h -diagonalisable) with finite-dimensional weight spaces are exhausted by

- The $(\lambda + 1)$ -dimensional modules $\mathcal{F}_\lambda$, with $\lambda \in \mathbb{Z}_{\geq 0}$. They are both highest and lowest weight. Weights: $\lambda, \lambda - 2, \dots, 2 - \lambda, -\lambda$
- The infinite-dimensional highest weight modules $\mathcal{H}_\lambda$, with $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$. Weights: $\lambda, \lambda - 2, \lambda - 4, \dots$
- The infinite-dimensional lowest weight modules $\mathcal{L}_\lambda$, with $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\leq 0}$. Weights: $\dots, \lambda + 4, \lambda + 2, \lambda$
- The infinite-dimensional weight modules $\mathcal{W}_{\lambda;\Delta}$, with $\lambda, \Delta \in \mathbb{C}$ satisfying $4(k+2)\Delta \neq \mu(\mu+2)$ for any $\mu \in \lambda + 2\mathbb{Z}$. They are neither highest nor lowest weight and there are isomorphisms $\mathcal{W}_{\lambda;\Delta} \cong \mathcal{W}_{\lambda+2;\Delta}$. Δ is the eigenvalue of the quadratic Casimir. Weights: $\dots, 2 + \lambda, \lambda, \lambda - 2, \dots$

All weight spaces are 1 dimensional.

Affine $\mathfrak{sl}(2)$

For $\widehat{\mathfrak{sl}}(2)$ with standard $\mathfrak{sl}(2)$ generators e, h, f , the Lie bracket specialises to

$$\begin{aligned}[h_m, e_n] &= 2e_{m+n}, & [h_m, f_n] &= -2f_{m+n}, \\ [h_m, h_n] &= \delta_{m+n,0} \cdot m \cdot 2 \cdot k\mathbf{1}, & [e_m, f_n] &= h_{m+n} + \delta_{m+n,0} \cdot m \cdot k\mathbf{1}.\end{aligned}$$

The elements e_0, h_0, f_0 generate a copy of $\mathfrak{sl}(2)$ inside $\widehat{\mathfrak{sl}}(2)$.

We split the algebra into three subalgebras (not a triangular decomposition)

$\mathfrak{g}_+ = \text{span} \{e_m, h_m, f_m m \geq 1\},$	annihilation op
$\mathfrak{g}_- = \text{span} \{e_{-m}, h_{-m}, f_{-m} m \geq 1\},$	creation op
$\mathfrak{g}_0 = \text{span} \{e_0, h_0, f_0, k\mathbf{1}\} = \mathfrak{sl}(2) \oplus \mathbb{C}k\mathbf{1}.$	zero modes

$\widehat{\mathfrak{sl}}(2)$ modules induced from $\mathfrak{sl}(2)$ modules

From any $\mathfrak{sl}(2)$ -module M , we can construct an $\widehat{\mathfrak{sl}}(2)$ -module $\widehat{M}$ in the following way:

- $\mathfrak{g}_0$ acts on M in the natural way.
- $\mathfrak{g}_+$ acts trivially.
- $\mathfrak{g}_-$ acts freely.

Example: Let $\mathbb{C} \cong \mathcal{F}_0$ be the trivial representation of $\mathfrak{sl}(2)$ with basis vector Ω . The first few grades of $V = \widehat{\mathbb{C}}$ are

h_0 -weight:	4	2	0	-2	-4
grade 0	Ω				
grade 1	$e_{-1}\Omega \quad h_{-1}\Omega \quad f_{-1}\Omega$				
grade 2	$e_{-1}^2\Omega$	$e_{-2}\Omega$ $h_{-1}e_{-1}\Omega$	$h_{-2}\Omega$ $h_{-1}^2\Omega$ $f_{-1}e_{-1}\Omega$	$f_{-2}\Omega$ $f_{-1}h_{-1}\Omega$	$f_{-1}^2\Omega$

Conformal field theories with Lie algebra symmetry

The induced module $V = \hat{\mathbb{C}}$ carries the structure of a vertex operator algebra (a.k.a. a chiral algebra).

Vertex operator algebra data

An $\mathbb{N}$ -graded $\mathbb{C}$ -vector space $V = \bigoplus_{n \geq 0} V[n]$, with an injective linear map

$$Y : V[n] \rightarrow \text{End}(V)[[z, z^{-1}]]$$
$$a \mapsto Y(a, z) = a(z) = \sum_{m \in \mathbb{Z}} a_m z^{-m-n},$$

where $a_m : V[l] \rightarrow V[l - m]$, $\forall l$, such that $\Omega \in V[0]$, $T \in V[2]$ and

$$Y(\Omega, z) = \mathbf{1}, \quad Y(T, z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}, \quad \text{span}\{L_n\}_{n \in \mathbb{Z}} = \text{Virasoro algebra}.$$

The vertex operator algebra data are subject to a long list of axioms that are not important here.

Conformal field theories with Lie algebra symmetry

Proposition

The affine module $V \cong \widehat{\mathbb{C}}$ induced from the trivial $\mathfrak{sl}(2)$ -representation is a vertex operator algebra with field map

$$e_{-1}\Omega \mapsto e(z) = \sum_{n \in \mathbb{Z}} e_n z^{-n-1}, \quad h_{-1}\Omega \mapsto h(z) = \sum_{n \in \mathbb{Z}} h_n z^{-n-1},$$

$$f_{-1}\Omega \mapsto f(z) = \sum_{n \in \mathbb{Z}} f_n z^{-n-1},$$

$$\begin{aligned} & Y(f_{-l_1} \cdots f_{-l_i} h_{-m_1} \cdots h_{-m_j} e_{-n_1} \cdots e_{-n_k} \Omega, z) \\ &= : \frac{\partial^{l_1-1} f(z)}{(l_1-1)!} \cdots \frac{\partial^{l_i-1} f(z)}{(l_i-1)!} \frac{\partial^{m_1-1} h(z)}{(m_1-1)!} \cdots \frac{\partial^{m_j-1} h(z)}{(m_j-1)!} \frac{\partial^{n_1-1} e(z)}{(n_1-1)!} \cdots \frac{\partial^{n_k-1} e(z)}{(n_k-1)!} :, \\ & T = \frac{1}{2(k+2)} \left(\frac{1}{2} h_{-1}^2 + e_{-1} f_{-1} + f_{-1} e_{-1} \right) \Omega \end{aligned}$$

Back to $\widehat{\mathfrak{sl}}(2)$ representation theory

- Even if the $\mathfrak{sl}(2)$ module one induces from is simple, the induced module need not be simple.
- Simplicity depends on the value of the level k .
- The levels $k = \frac{u}{v} - 2$, $u \in \mathbb{Z}_{\geq 2}$, $v \in \mathbb{Z}_{\geq 1}$, $\gcd(u, v) = 1$, are called **admissible**.

Proposition [Kac, Kazhdan]

The induction of the trivial module, $V = \widehat{\mathbb{C}}$, is non-simple if and only if k is admissible or $k = -2$. At admissible levels, there exists a unique submodule. It is generated by a highest weight vector of $\mathfrak{sl}(2)$ -weight $2(u - 1)$ at grade $(u - 1)v$.

$\mathfrak{sl}(2)$ minimal models

The $\mathfrak{sl}(2)$ minimal model vertex operator algebras $A(u, v)$ are the simple quotients of the induced module $V = \widehat{\mathbb{C}}$ at admissible levels.

Corollary

The modules of $A(u, v)$ are those of the vertex operator algebra $V = \widehat{\mathbb{C}}$ on which the fields associated to vectors in the submodule of $V = \widehat{\mathbb{C}}$ act trivially.

Proposal for an $A(u, v)$ module category:

$\widehat{\mathfrak{sl}}(2)$ modules that are induced from $\mathfrak{sl}(2)$ weight modules which are annihilated by the submodule of $V = \widehat{\mathbb{C}}$ (also quotients and submodules).

Example: non-negative integer k

Proposition [Malikov-Feigin-Fuks]

If k is a non-negative integer, then the highest weight vector that generates the submodule of $V = \widehat{\mathbb{C}}$ and the corresponding field are

$$e_{-1}^{k+1}\Omega \leftrightarrow Y(e_{-1}^{k+1}\Omega, z) = :e(z)^{k+1} :.$$

Let M be an $\mathfrak{sl}(2)$ module and $\widehat{M}$ its induction. On an element $m \in M \subset \widehat{M}$ the action of $:e(z)^{k+1}:$ expands as

$$:e(z)^{k+1}:m = e_0^{k+1}mz^{-k-1} + \mathcal{O}(z^{-k}).$$

Thus, $\widehat{M}$ can only be an $A(k+2, 1)$ module if e_0^{k+1} acts trivially on M .

Proposition

The simple $A(k+2, 1)$ modules are the simple quotients induced from $\mathcal{F}_\lambda$, $\lambda = 0, \dots, k$, the $\lambda + 1$ -dimensional simple $\mathfrak{sl}(2)$ modules.

General admissible levels

At general admissible level the highest weight vector in $\widehat{\mathbb{C}}$ is very complicated, but the same analysis can be performed. Let

$$\lambda_{r,s} = r - 1 - s \frac{u}{v}, \quad \Delta_{r,s} = \frac{r^2 - 1}{2} + \frac{s^2 u^2}{2 v^2} - rs \frac{u}{v}.$$

Proposition [Adamović, Milas] [Ridout, SW]

The simple $A(u, v)$ modules are exhausted by the following:

- The simple quotients induced from the finite-dimensional modules $\mathcal{F}_r$, where $0 \leq r \leq u - 2$.
- The simple quotients induced from the infinite-dimensional highest weight modules $\mathcal{H}_{\lambda_{r,s}}$, where $1 \leq r \leq u - 1$ and $1 \leq s \leq v - 1$.
- The simple quotients induced from the infinite-dimensional lowest weight modules $\mathcal{L}_{-\lambda_{r,s}}$, where $1 \leq r \leq u - 1$ and $1 \leq s \leq v - 1$.
- The simple quotients induced from the infinite-dimensional weight modules $\mathcal{W}_{\lambda; \Delta_{r,s}}$, where $1 \leq r \leq u - 1$ and $1 \leq s \leq v - 1$, $4(k+2)\Delta_{r,s} \neq \mu(\mu+2)$ for all $\mu \in \lambda + 2\mathbb{Z}$ and $\mathcal{W}_{\lambda; \Delta_{r,s}} \cong \mathcal{W}_{\lambda; \Delta_{u-r, v-s}}$.

Spectral flow

The previous list of modules does not close under fusion or modular transformations. This is because we have made a rather arbitrary choice of $\mathfrak{sl}(2)$ in $\widehat{\mathfrak{sl}}(2)$. There are many more choices that are all related by an algebra automorphism σ called spectral flow.

$$\begin{aligned}\sigma(e_m) &= e_{m-1}, & \sigma(f_m) &= f_{m+1}, \\ \sigma(h_m) &= h_m - \delta_{m,0}k\mathbf{1}, & \sigma(k\mathbf{1}) &= k\mathbf{1}.\end{aligned}$$

This automorphism tilts our decomposition of $\widehat{\mathfrak{sl}}(2)$

$$\begin{array}{lll} e_2 & h_2 & f_2 \\ e_1 & h_1 & f_1 \\ e_0 & h_0, k\mathbf{1} & f_0 \\ e_{-1} & h_{-1} & f_{-1} \\ e_{-2} & h_{-2} & f_{-2} \end{array} \quad \begin{array}{l} \sigma^0(\mathfrak{g}_+) \\ \sigma^0(\mathfrak{g}_0) \\ \sigma^0(\mathfrak{g}_-) \end{array}$$

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Spectral flow of modules

The algebra automorphism σ induces a homomorphism of modules which tilts gradings. Example: $\sigma : \widehat{\mathbb{C}} \rightarrow \sigma(\widehat{\mathbb{C}})$

$\widehat{\mathbb{C}}$:

h_0 -weight:	4	2	0	-2	-4
grade 0	Ω				
grade 1	$e_{-1}\Omega \quad h_{-1}\Omega \quad f_{-1}\Omega$				
grade 2	$e_{-1}^2\Omega$	$e_{-2}\Omega$ $h_{-1}e_{-1}\Omega$	$h_{-2}\Omega$ $h_{-1}^2\Omega$ $f_{-1}e_{-1}\Omega$	$f_{-2}\Omega$ $f_{-1}h_{-1}\Omega$	$f_{-1}^2\Omega$

$\sigma(\widehat{\mathbb{C}})$:

h_0 -weight:	$k+4$	$k+2$	k	$k-2$
grade 0	$\sigma(e_{-1}^2\Omega)$	$\sigma(e_{-1}\Omega)$	$\sigma(\Omega)$	
grade 1		$\sigma(e_{-2}\Omega)$ $\sigma(h_{-1}e_{-1}\Omega)$	$\sigma(h_{-1}\Omega)$	
grade 2			$\sigma(h_{-2}\Omega)$ $\sigma(h_{-1}^2\Omega)$ $\sigma(f_{-1}e_{-1}\Omega)$	$\sigma(f_{-1}\Omega)$

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$\widehat{\mathbb{C}}$:

h_0 -weight:	4	2	0	-2	-4
grade 0	Ω				
grade 1	$e_{-1}\Omega \quad h_{-1}\Omega \quad f_{-1}\Omega$				
grade 2	$e_{-1}^2\Omega$	$e_{-2}\Omega$ $h_{-1}e_{-1}\Omega$	$h_{-2}\Omega$ $h_{-1}^2\Omega$ $f_{-1}e_{-1}\Omega$	$f_{-2}\Omega$ $f_{-1}h_{-1}\Omega$	$f_{-1}^2\Omega$

$\sigma(\widehat{\mathbb{C}})$:

h_0 -weight:	$k+4$	$k+2$	k	$k-2$
grade 0	$e_0^2\sigma(\Omega)$	$e_0\sigma(\Omega)$	$\sigma(\Omega)$	
grade 1		$e_{-1}\sigma(\Omega)$ $h_{-1}e_0\sigma(\Omega)$	$h_{-1}\sigma(\Omega)$	
grade 2			$h_{-2}\sigma(\Omega)$ $h_{-1}^2\sigma(\Omega)$ $f_{-2}e_0\sigma(\Omega)$	$f_{-2}\sigma(\Omega)$

The correct module category for conformal field theory

If we take the simple $A(u, v)$ modules classified previously, their spectral flows to all orders and allow extensions, then:

- 1 The action of the modular group closes on characters.
- 2 Fusion is believed to close.

Open problem: Find a nicer characterisation of this category.

Outlook

All of the ideas presented today generalise nicely to higher rank Lie algebras. Nice future problems include:

- 1 The induction of the trivial module is generically simple and contains a unique submodule generated by a highest weight vector at admissible levels.
- 2 Identify admissible modules among induced modules.
- 3 Check that the modular group closes on the characters of spectrally flowed modules.
- 4 Check that fusion closes on spectrally flowed modules and their extensions.
- 5 Find a nice way to characterise the module category.